HOMOTHETIES BY ELEMENTS OF THE MULTIPLICATIVE SET ARE UNIVERSALLY INJECTIVE, AND SO IS THE LOCALIZATION MAP, OVER A LOCAL ARTINIAN-FLAVOURED FLAT BASE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Exactness of the multiplication sequence and flatness of the quotient N/sN. Algebra, label algebra-lemma-mod-injective-general (invoked at proof line 1838). Under the local homomorphism and finite presentation hypotheses it gives the short exact sequence $0 \to N \to N \to N/sN \to 0$ with N/sN flat over R.
- Flatness of the quotient as a criterion for universal injectivity. Algebra, label algebra-lemma-flat-tor-zero (invoked at proof line 1842). It converts the flatness of N/sN into universal injectivity of $s : N \to N$ over R.
- The spectrum of the homotheties and the notation Σ, N used in the statement. Same chapter flat.tex, label lemma-homothety-spectrum (statement line 1823). It fixes the multiplicative subset Σ and the module N to which the extra hypotheses of this lemma are added.

Source reference. Original lemma 00HL: sources/stacks-chow/tags.txt, source lines 647–647. Navigation pointer for \ref{algebra-lemma-flat-tor-zero} used inside the counted span (line 1842); the argument that consumes it is retained in the span.

Source reference. Original lemma 046Y: sources/stacks-chow/tags.txt, source lines 5489–5489. Navigation pointer for \ref{algebra-lemma-mod-injective-general} used inside the counted span (line 1838); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DE: sources/stacks-flat/pinned-original-source.tex, source lines 1783–1783. Navigation pointer for \ref{lemma-homothety-spectrum}

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used inside the counted span (line 1823); the argument that consumes it is retained in the span.

Lemma 0.1. *Assumption and notation as in Lemma 05DE. Assume moreover that*

(1) *S is local and $R \to S$ is a local homomorphism,*
(2) *S is essentially of finite presentation over R,*
(3) *N is finitely presented over S, and*
(4) *N is flat over R.*

Then each $s \in \Sigma$ defines a universally injective R-module map $s : N \to N$, and the map $N \to \Sigma^{-1}N$ is R-universally injective.

Proof. By Algebra, Lemma 046Y the sequence $0 \to N \to N \to N/sN \to 0$ is exact and N/sN is flat over R. This implies that $s : N \to N$ is universally injective, see Algebra, Lemma 00HL. The map $N \to \Sigma^{-1}N$ is universally injective as the directed colimit of the maps $s : N \to N$. □